Stochastic Attention via Langevin Dynamics on the Modern Hopfield Energy

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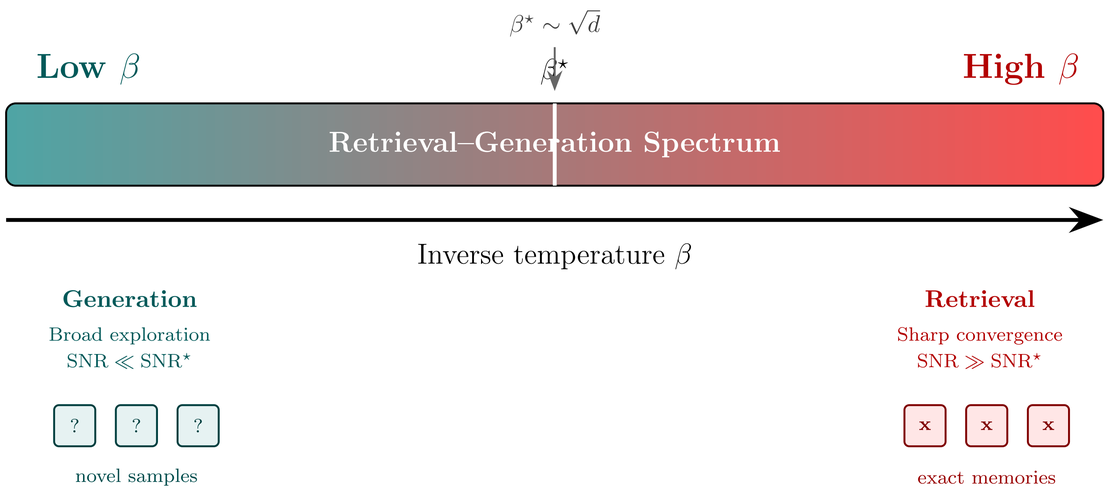
Motivation

Attention heads are deterministic: the same query always returns the same softmax-weighted memory average. Yet many tasks require *generating* novel, plausible patterns from structured memory.
We show that the gradient of the **modern Hopfield energy** is exactly the identity minus the attention map, and that applying **Langevin dynamics** to this energy yields *stochastic attention*: a **training-free sampler** controlled by a single inverse temperature β.



The critical temperature $\beta^\star$ marks the retrieval–generation boundary, identified analytically via an entropy inflection condition: $\beta^\star \sim \sqrt{d}$ for random unit-norm memories. **No score network, no training loop, no contrastive objective**:the score is the attention map itself.

Method: Stochastic Attention Update

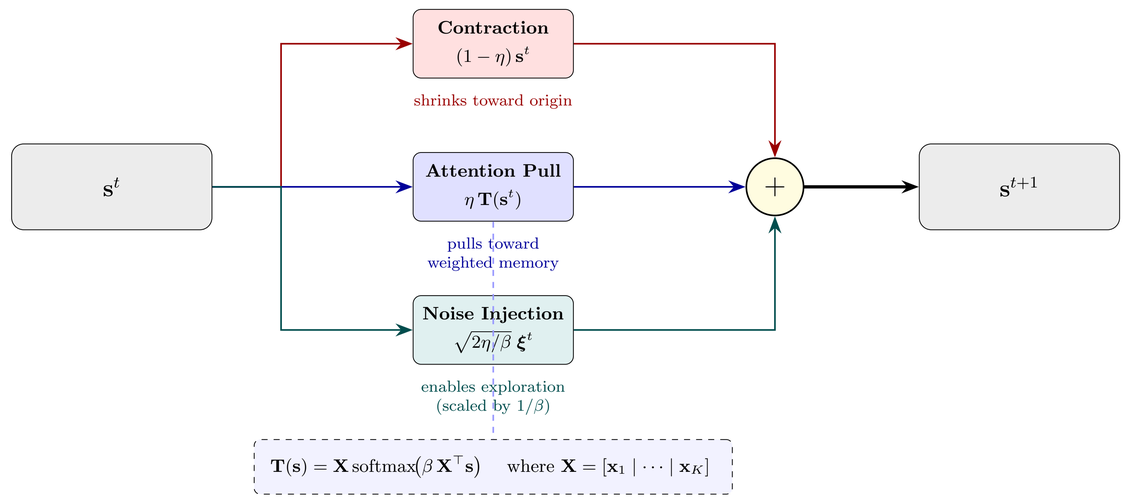
$\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{N\times K}$ stores K patterns. To generate, we sample from $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$.
The ULA turns any smooth energy U into a sampler:

$$\boldsymbol{\xi}_{t+1} = \boldsymbol{\xi}_t - \tilde{\alpha}\,\nabla U(\boldsymbol{\xi}_t) + \sqrt{2\tilde{\alpha}}\,\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

The gradient of the modern Hopfield energy is exactly the attention residual:

$$\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \underbrace{\mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top \boldsymbol{\xi})}_{\mathbf{T}(\boldsymbol{\xi})\,=\,\text{softmax attention map}}.$$

Substituting ∇E into the ULA (with $U = \beta E$, $\tilde{\alpha} = \alpha/\beta$):

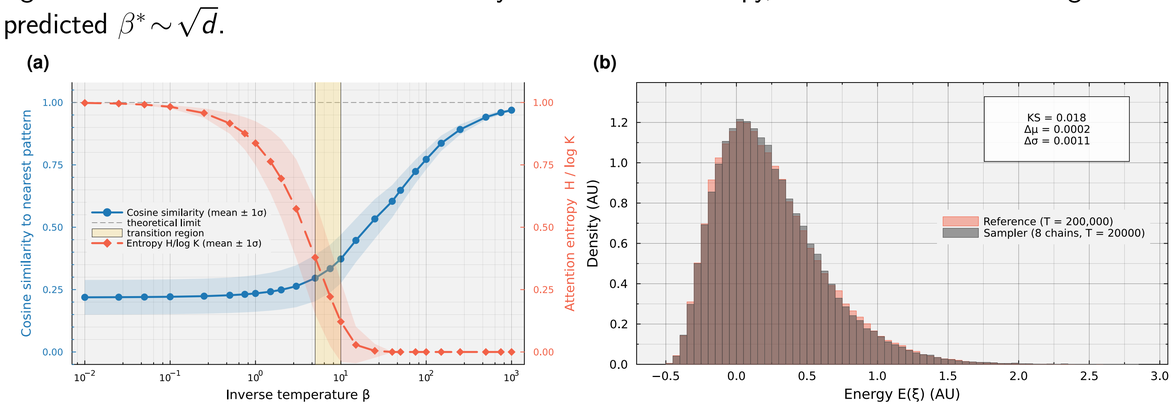


$$\boldsymbol{\xi}_{t+1} = \underbrace{(1-\alpha)\,\boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha\,\mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\tfrac{2\alpha}{\beta}}\,\boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}$$

Step cost: $\mathcal{O}(dK)$: same as one attention head; no training required.
Convergence: W_2-geometric in the convex regime ($\beta\sigma_{\max}^2 < 2$); MALA acceptance = 99.2% at $\alpha = 0.01$ confirms negligible discretization bias.
Temperature selection: β^* via entropy inflection $d^2H/d\beta^2 = 0$; scales as $\beta^* \sim \sqrt{d}$.

Experiment 1:Phase Behavior & Convergence (Synthetic)

Does β really act as a continuous order parameter between generation and retrieval, and does the sampler converge to the correct Boltzmann target? If the theory is right, we should see a smooth sigmoidal transition in both cosine similarity and attention entropy, with the inflection landing near the predicted $\beta^* \sim \sqrt{d}$.



(a) $d = 64$, $K = 16$. Cosine similarity (blue) and scaled entropy (coral) exhibit a smooth sigmoidal transition centered at $\beta \approx 5$–10 (gold band), consistent with the predicted $\beta^* \sim \sqrt{d}$.

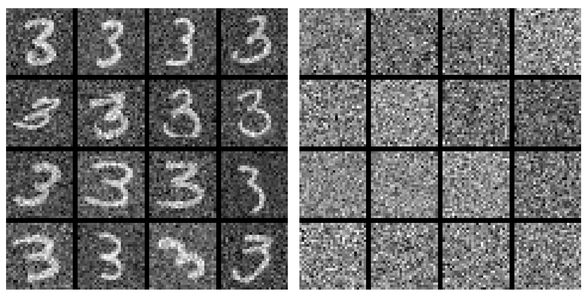
(b) $d = 8$, $K = 4$, $\beta = 5$. Eight independent chains (gray) converge to a long-run reference (coral); KS statistic confirms the correct Boltzmann target.

Experiment 2:MNIST Image Generation ($K = 100$, $d = 784$)

Can stochastic attention generate structured, recognizable images from a small memory, and how does it compare to learned generative models on the same data?
$K = 100$ images of digit "3"; 30 independent chains; 150 generated samples. $\beta = 2000$ (retrieval) and $\beta = 200$ (generation, near β^*).

Method	Nov.↑	Div.↑	Energy↓
Bootstrap	0.000	0.459	−0.500
Gaussian perturb.	0.004	0.450	−0.496
Convex combo	0.092	0.008	−0.399
GMM-PCA	0.198	0.419	−0.303
VAE (latent = 8)	0.214	0.441	−0.286
DDPM†	0.938	0.991	+0.44
MALA ($\beta = 2000$)	0.151	0.598	−0.305
SA ($\beta = 2000$)	0.152	0.600	−0.303
SA ($\beta = 200$)	**0.548**	**0.885**	+1.467

†DDPM generates isotropic noise despite 5,000 training epochs.



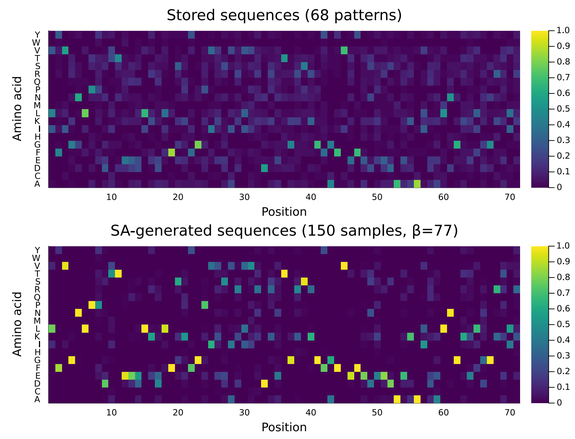
SA ($\beta = 2000$) **DDPM**

SA at $\beta = 200$ achieves **2.6**× the novelty and **2.0**× the diversity of the VAE, while matching the Metropolis-corrected MALA gold standard. Crucially, DDPM fails entirely despite 5,000 training epochs — the training-free score function provides a structural advantage that a learned model cannot recover from only 100 examples.

Experiment 3:Protein Sequence Generation (Pfam RRM, $K = 68$, $d = 59$)

Does the training-free score function preserve domain-level biological fidelity when generating novel sequences, something learned models struggle with at small K?
68 aligned RRM sequences; one-hot encoded, PCA-reduced (95% variance). Entropy inflection gives $\beta^* = 3.85$ (half the random-pattern prediction $\sqrt{d} = 7.68$; conserved residues increase effective similarity variance).

Method	Nov.↑	Div.↑	SeqID	KL↓
Bootstrap	0.000	1.000	0.645	0.133
VAE	0.621	0.834	0.532	0.416
SA (retrieval)	0.243	0.991	0.616	0.107
SA (gen.)	**0.623**	**1.001**	0.538	**0.060**

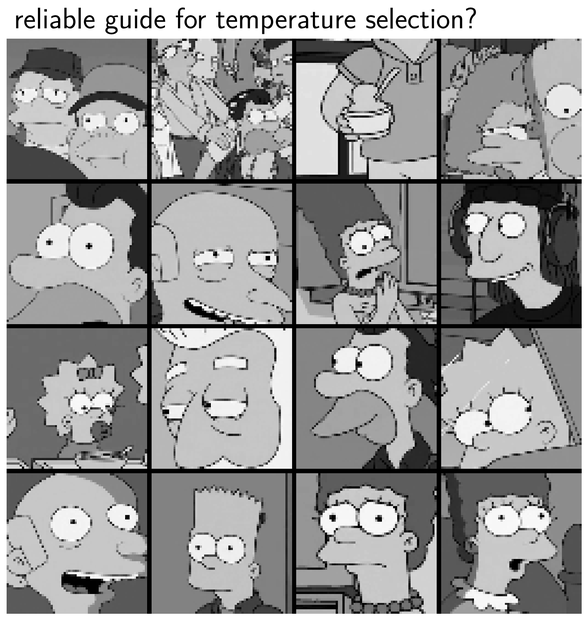


Stored (top) vs. SA-generated (bottom) amino acid frequencies.

SA matches the VAE on novelty while achieving **6.9**× lower amino acid composition divergence (KL = 0.060 vs. 0.416). Because the Langevin gradient is recomputed from the stored sequences at every step, it continuously steers samples back toward the family's compositional profile — fidelity the VAE loses when trained on only 68 sequences.

Experiment 4:Scaling to Face Images — Simpsons ($d = 4{,}096$)

Does SA still work when we scale to high-dimensional image data, and does the $\beta^* \sim \sqrt{d}$ rule remain a reliable guide for temperature selection?



SA-generated faces ($\beta = 10{,}000$, retrieval regime).

64×64 grayscale Simpsons character images ($d = 4{,}096$). The SNR rule prescribes $\beta^* \sim \sqrt{4096} = 64$; we run at $\beta = 10{,}000$ (well inside retrieval).
SA achieves novelty 0.159±0.000 and diversity 0.293±0.001, with **identical method ranking to MNIST** across a **5.2**× increase in dimension. The SNR selection rule transfers directly from 784 to 4,096 dimensions with no tuning.

Limitations & Future Work

ULA bias scales with step size α; MALA (with Metropolis correction) is preferred at larger α.
Inter-basin mixing at high β is exponentially slow; the multi-chain protocol provides a practical workaround, but quantitative scaling laws remain open.
Fixed memory: X must be known at sampling time; streaming or latent memory settings require extension.
Generation fidelity at high temperature yields structured interpolations rather than crisp samples; operating near β^* improves quality.
Future: multi-head attention, hierarchical memories, retrieval-augmented generation, in-context learning.

github.com/varnerlab/stochastic-attention-study-paper